1. Optional Questions (Multiple Choice)

(Each question has four options, with one correct answer)

1. **Which model is primarily used for text generation in Generative AI?**
   - A) CNN
   - B) RNN
   - C) GAN
   - D) Transformer
2. **What is the key difference between Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs)?**
   - A) VAEs use adversarial training, while GANs do not
   - B) GANs use a generator-discriminator setup, VAEs do not
   - C) VAEs generate images, while GANs cannot
   - D) Both are completely identical in architecture
3. **In a GAN, what is the primary goal of the discriminator?**
   - A) Generate realistic images
   - B) Identify fake data from real data
   - C) Encode features into latent space
   - D) Reduce the model size
4. **Which activation function is commonly used in GANs for stable training?**
   - A) ReLU
   - B) Sigmoid
   - C) Leaky ReLU
   - D) Softmax
5. **What does "zero-shot learning" mean in the context of Generative AI?**
   - A) Model requires no training
   - B) Model generalizes to new tasks without prior training data
   - C) Model learns after a single sample
   - D) Model generates synthetic data
6. **Which dataset is most commonly used to train AI models for text generation?**
   - A) CIFAR-10
   - B) MNIST
   - C) ImageNet
   - D) Common Crawl
7. **What is the role of the "temperature" parameter in text generation using a Transformer model?**
   - A) Controls randomness in output

- B) Increases model speed
- C) Reduces model size
- D) Controls GPU usage

8. What does "latent space" refer to in Generative AI?

- A) A hidden feature representation of data
- B) A compression technique
- C) A specific AI algorithm
- D) A storage technique

9. Which loss function is commonly used for training GANs?

- A) Mean Squared Error
- B) Binary Cross Entropy
- C) Categorical Cross Entropy
- D) KL Divergence

10. What is a common issue faced when training GANs?

- A) Vanishing gradients
- B) Mode collapse
- C) Overfitting
- D) Underfitting

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2. Fill in the Blank Questions

11. The architecture of GAN consists of a _____ and a discriminator.
12. A Transformer model relies heavily on the _____ mechanism for processing sequential data.
13. The _____ loss function is often used in Variational Autoencoders to optimize latent space distribution.
14. GPT models use the _____ network to generate coherent text responses.
15. The primary challenge in training deep generative models is _____.
16. Stable Diffusion is an example of a _____ model used for generating images.
17. In GANs, the generator tries to _____ the discriminator.
18. The _____ technique is used to generate synthetic data from limited datasets.
19. Autoencoders are primarily used for _____ learning.
20. Reinforcement Learning with Human Feedback (RLHF) is used in models like _____ for better alignment with human preferences.

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3. One-Liner Questions

21. What does "attention" mean in the Transformer model?
22. Name one major advantage of diffusion models over GANs.
23. What is the role of the KL divergence in VAEs?
24. How do large language models improve contextual understanding?
25. Why is temperature tuning important in text generation?
26. What is mode collapse in GANs?
27. Name a generative AI model used for music generation.
28. What is an embedding in the context of NLP?
29. What is the purpose of latent variables in VAEs?
30. How does contrastive learning help in representation learning?

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4. Theory-Based Questions

31. Explain the difference between a Variational Autoencoder (VAE) and a Generative Adversarial Network (GAN).
32. How do diffusion models work in image generation?
33. What are the ethical concerns of using AI-generated content?
34. Explain the role of self-attention in Transformers.
35. How does reinforcement learning improve generative AI models?
36. What are the common evaluation metrics for generative AI models?
37. Describe the importance of large-scale datasets in training generative models.
38. How does conditional GAN (cGAN) differ from traditional GAN?
39. Explain the significance of loss balancing in GAN training.
40. How does transfer learning help in improving generative models?

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5. Coding-Based Questions

41. Write a Python function to generate synthetic text using OpenAI's GPT-3 API.
42. Implement a simple GAN architecture using TensorFlow/Keras.
43. Write a PyTorch-based VAE model for image generation.
44. Write a Python function to preprocess text for training a generative AI model.
45. Implement a Transformer block in PyTorch.
46. Write a script to fine-tune a GPT-2 model using Hugging Face Transformers.
47. Generate an image using Stable Diffusion API in Python.
48. Implement a text summarizer using a Transformer model.
49. Write a function to evaluate the perplexity of a language model.
50. Create a GAN-based face generator using PyTorch.